

Mutual Fund AnalyticsMutual Fund Analytics **Dashboard**

Internship Project Report

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Technologies Used

Python • Pandas • SQLite • SQL • Streamlit • Plotly • Git • GitHub

1. Introduction

The Mutual Fund Analytics Dashboard is an end-to-end data analytics project that analyzes mutual fund performance using Python, SQL, SQLite, and Streamlit. The project demonstrates the complete analytics lifecycle, from data ingestion and preprocessing to database management, visualization, and dashboard development.

The objective was to build an interactive dashboard that helps users compare mutual funds based on performance, risk, expense ratio, assets under management (AUM), and other important financial metrics.

2. Project Objectives

The primary objectives of this project were:

- Import and organize mutual fund datasets.
 - Perform data cleaning and exploratory data analysis.
 - Store processed data in an SQLite database.
 - Write SQL queries to analyze fund performance.
 - Create SQL views for dashboard-ready datasets.
 - Build an interactive dashboard using Streamlit.
 - Generate insights for investment analysis.
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3. Dataset Description

The project consists of multiple datasets covering different aspects of mutual funds.

Main Datasets

- Fund Master
- NAV History
- Scheme Performance
- Assets Under Management (AUM)
- Monthly SIP Inflows
- Category Inflows
- Industry Folio Count
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices

These datasets include information such as:

- Scheme Name
 - AMFI Code
 - Fund House
 - Category
 - Risk Level
 - Returns (1-Year, 3-Year, 5-Year)
 - Expense Ratio
 - Morningstar Rating
 - Net Asset Value (NAV)
 - Assets Under Management (AUM)
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4. Tools and Technologies

Tool	Purpose
Python	Data Processing
Pandas	Data Analysis
SQLite	Database
SQL	Data Querying
Plotly	Interactive Charts
Streamlit	Dashboard Development
Git	Version Control
GitHub	Project Hosting
VS Code	Development Environment

5. Data Analysis

The datasets were analyzed using Python and Pandas.

The following checks were performed:

- Missing value analysis
- Duplicate record detection
- Data type validation

- Summary statistics
- Category distribution
- Fund house analysis
- Expense ratio analysis
- Return analysis
- Risk analysis

Python scripts were developed separately for each dataset to keep the project modular and maintainable.

6. Database Design

All processed datasets were imported into an SQLite database.

Database Tables

- fund_master
- nav_history
- scheme_performance
- aum_by_fund_house
- monthly_sip_inflows
- category_inflows
- industry_folio_count
- investor_transactions
- portfolio_holdings
- benchmark_indices

SQL views were created to simplify dashboard development:

- performance_summary
 - latest_nav
 - dashboard_dataset
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7. SQL Analysis

Several SQL queries were written to generate business insights.

Examples include:

- Top-performing mutual funds
- Highest Assets Under Management
- Average expense ratio

- Risk category distribution
- Morningstar rating analysis
- Fund house comparison
- Category-wise performance
- Sharpe ratio comparison

These SQL queries provided the foundation for the dashboard.

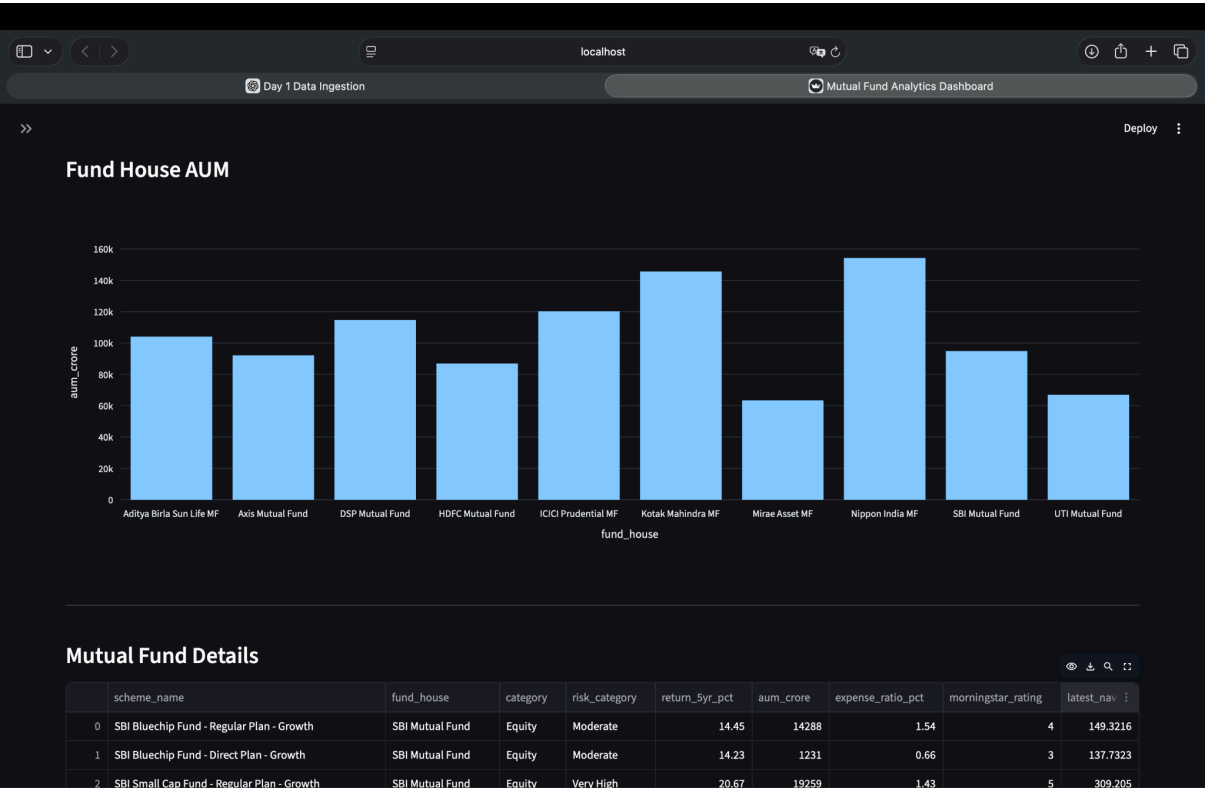
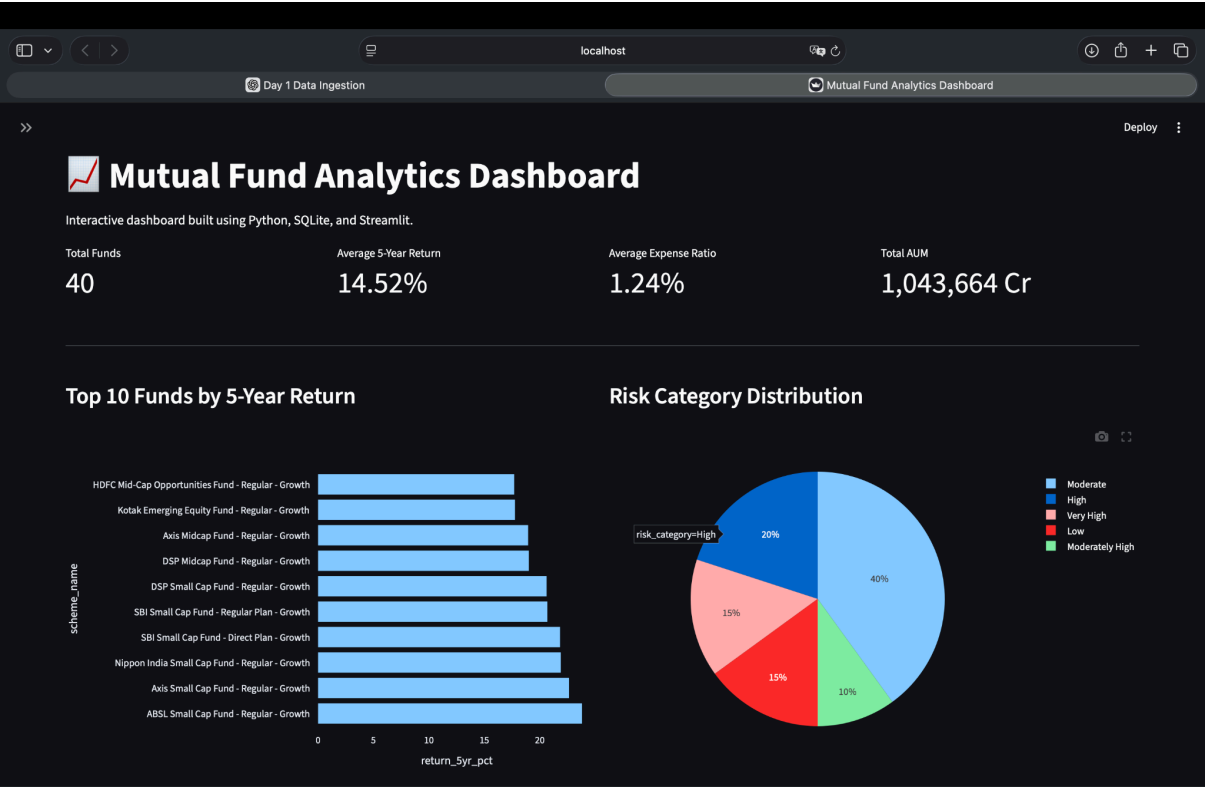
8. Dashboard Development

An interactive dashboard was developed using Streamlit.

Dashboard Features

- KPI cards
- Category filter
- Fund house filter
- Scheme search
- Top 10 performers
- Risk distribution chart
- Fund House AUM comparison
- Interactive data table
- Download filtered data

The dashboard enables users to explore the data interactively without writing SQL queries.



The screenshot shows a Streamlit dashboard with a sidebar containing navigation links for various data ingestion and analysis tasks. The main area displays a table titled 'Mutual Fund Details' with the following data:

	scheme_name	fund_house	category	risk_category	return_5yr_pct	aum_crore	expense_ratio_pct	morningstar_rating	latest_nav
0	SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund	Equity	Moderate	14.45	14288	1.54	4	149.3216
1	SBI Bluechip Fund - Direct Plan - Growth	SBI Mutual Fund	Equity	Moderate	14.23	1231	0.66	3	137.7323
2	SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	Equity	Very High	20.67	19259	1.43	5	309.205
3	SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	Equity	Very High	21.82	36061	0.72	4	105.4849
4	SBI Magnum Gilt Fund - Regular Plan - Growth	SBI Mutual Fund	Debt	Low	5.43	24101	0.77	5	54.2038
5	HDFC Top 100 Fund - Regular Plan - Growth	HDFC Mutual Fund	Equity	Moderate	11.32	6434	1.55	5	583.6113
6	HDFC Top 100 Fund - Direct Plan - Growth	HDFC Mutual Fund	Equity	Moderate	13.48	10611	0.92	4	1204.9571
7	HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	Equity	High	17.69	23185	1.38	5	342.0072
8	HDFC Mid-Cap Opportunities Fund - Direct - Growth	HDFC Mutual Fund	Equity	High	15.85	18792	0.78	4	188.3619
9	HDFC Short Term Debt Fund - Regular - Growth	HDFC Mutual Fund	Debt	Low	6.41	27953	0.56	3	31.8843

9. Key Insights

The analysis identified several important observations:

- Equity funds generated higher long-term returns than debt funds.
- Small-cap funds delivered the highest five-year returns.
- Moderate-risk funds represented the largest proportion of schemes.
- Expense ratios varied significantly across fund categories.
- Nippon India Mutual Fund managed the largest total AUM in the dataset.
- Morningstar ratings were concentrated between 4 and 5 stars.

10. Challenges Faced

During development, several technical challenges were encountered:

- Configuring the Python virtual environment
- Loading datasets into SQLite
- Creating SQL views correctly
- Exporting dashboard datasets
- Building the Streamlit dashboard

- Managing project folder structure

These challenges were resolved through debugging and iterative testing.

11. Learning Outcomes

This project strengthened practical skills in:

- Python programming
 - Data cleaning
 - Exploratory Data Analysis
 - SQL query writing
 - Database management
 - Dashboard development
 - Data visualization
 - Git and GitHub workflows
 - End-to-end analytics project development
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12. Future Enhancements

Potential improvements include:

- Integration with live NAV APIs
 - Machine learning-based return prediction
 - Portfolio optimization tools
 - Investor recommendation engine
 - Cloud deployment of the Streamlit dashboard
 - Authentication for user-specific dashboards
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13. Conclusion

The Mutual Fund Analytics Dashboard successfully demonstrates an end-to-end analytics workflow. The project integrates data processing, SQL, database management, and interactive visualization into a single application.

The dashboard enables users to analyze mutual fund performance efficiently and provides meaningful insights into fund returns, risk, AUM, and expense ratios. It also showcases practical skills that are directly applicable to real-world data analytics and business intelligence projects.

Appendix

Project Folder Structure

```
mf-analytics-project/  
├── dashboard/  
├── database/  
├── data/  
├── reports/  
├── scripts/  
├── sql/  
├── README.md  
├── requirements.txt  
└── .gitignore
```

Deliverables

- Python Analysis Scripts
 - SQLite Database
 - SQL Queries
 - SQL Views
 - Dashboard
 - README Documentation
 - Internship Report
 - GitHub Repository
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